

Lakshmi Ramesh

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Research interests

High dimensional statistics - Non-regular log-concave location families, **Algebraic statistics** - Sampling points on semi-algebraic sets, **Convex geometry** - Polyhedral convex geometry, volumes of convex bodies, probabilistic convex geometry, **Computer algebra** - Volume computation, random sampling algorithms, free resolution computations, parallelisation.

Education

- Universität Bielefeld** - *Doctoral candidate* 2024 - 2027 (tentative)
Supervised by Prof. Martin Wahl and Prof. Bernd Sturmfels
- RPTU Kaiserslautern** - *Master of Science* 2021 - 2024
Mathematics with specialisation in Algebra, Geometry and Computer Algebra
- Indian Institute of Science Education and Research Mohali** - *BS-MS Dual degree* 2015 - 2020
Major in Mathematics

Publications and Preprints

- Schreyer Resolutions over the Exterior Algebra** September 2026
Janko Boehm, Lakshmi Ramesh
arXiv 2609.37610
- Exact volumes of semi-algebraic convex bodies** July 2026
Lakshmi Ramesh, Nicolas Weiss
ISSAC '26: Proceedings of the 2026 International Symposium on Symbolic and Algebraic Computation
- Points on rational curves and the ABCT variety** May 2025
Daniele Agostini, Lakshmi Ramesh, Dawei Shen
Le Matematiche vol. 80 (2025), 103-122
- Invariant Theory** January 2025
Wolfram Decker, Lakshmi Ramesh, Johannes Schmidt
Book chapter (Springer)

Contributed talks

- Westfälische Stochastiktage** September 2026
Universität Paderborn
- European Women in Mathematics General Meeting 2026** September 2026
University of Warwick
- Young researchers' conference on combinatorial synergies 2026** May 2026
Free University Berlin
- Workshop Metric Algebraic Geometry: Starting Local** April 2026
Max Planck Institute for Mathematics in the Sciences
- Nonlinear Algebra Seminar** February 2026
Max Planck Institute for Mathematics in the Sciences

KOLKOM 2025
Universität Bielefeld

November 2025

Oberseminar Probability Theory and Mathematical Statistics
Universität Bielefeld

October 2025

Cohomology Theories of Automorphic Forms
Mathematisches Forschungsinstitut Oberwolfach

August 2025

TRR 358 Bielefeld Graduate Seminar
Universität Bielefeld

July 2025

Oberseminar Applied Algebra and Discrete Mathematics
Universität Osnabrück

June 2025

Combinatorial Synergies East
Max Planck Institute for Mathematics in the Sciences

March 2025

Non-linear Algebra Seminar
Max Planck Institute for Mathematics in the Sciences

March 2025

Computer Algebra Projects

Algorithms for Free Resolutions over the Exterior Algebra with Parallel Structures 2023 - 2024

Masters Thesis project supervised by Dr. Janko Boehm, RPTU Kaiserslautern

Using the Bernstein-Gelfand-Gelfand correspondence and the Tate resolution, one can compute the cohomology numbers of a coherent sheaf over project space. This requires computing the minimal free resolution of a module over the exterior algebra. I adapted Schreyer's new syzygy algorithm to the non-commutative setting of the exterior algebra. I have implemented the algorithm in Singular, and this can be parallelised using the Singular- GPI Space interface.

Invariant Rings 2023 - 2024

I use Harm Derksen's algorithm to compute fundamental invariants of linearly reductive groups and Cayley's Omega process for SL_n , and implemented invariant rings of reductive groups, as well as tori in OSCAR. This project was funded by the SFB-TRR 195 project under the supervision of Prof. Max Horn and Prof. Wolfram Decker.

Modules over Dedekind Domains 2022 - 2023

I developed functionality for modules over Dedekind domains in Hecke (dependency of OSCAR), including modules with torsion, quotient modules, module homomorphisms and elemental functionality. I also implemented a corrected version of Henry Cohen's algorithm for the Smith Normal Form, giving a canonical representation to a module over a Dedekind Domain. This project was funded by the SFB-TRR 195 project under the supervision of Prof. Claus Fieker.

Work Experience

Organisation of Reading Groups 2024- 2025

I organised two reading groups between the University of Bielefeld and the University of Osnabrück that met once a week to study "Metric Algebraic Geometry" by Paul Breiding, Kathlen Kohn, and Bernd Sturmfels, as well as "3254 and all that" by David Eisenbud and Joe Harris.

Research Assistant, RPTU Kaiserslautern 2022 - 2024

This position is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) through the SFB-TRR 195 project for 'Symbolic Tools in Mathematics and their Application'. I study algorithms in computer algebra related to algebraic number theory and algebraic geometry, and implement them in 'Open Source Computer Algebra Research system', OSCAR.

Tutor

2021

In this teaching role at *Tuition Wayanad Online*, I prepared course material in mathematics and physics. I taught lessons and handled tutorials for high school students from around India over video call (due to the pandemic). I gained skills in teaching and designing learning material.

Scholarships

INSPIRE Scholarship

2015 - 2020

Awarded by the Government of India

The INSPIRE scholarship, awarded to the top percentile of school graduates who pursue pure sciences, is a scheme by the Ministry of Science and Technology of the Government of India.